NOT MEASUREMENT SENSITIVE
MIL-P-85856/39
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS CHROMIUM CARBIDE PLUS NICKEL CHROMIUM ALLOY (FINE), TYPE III, COMPOSITION C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Percent by Weight		Components
Min.	Max.	
74.0	76.0	Type II, Composition B, Class 1
24.0	26.0	Type I, Composition U, Class 1

Sprayed Coating Properties

Bond Strength	34.5 MPa (5000 psi) min.
Hardness	450 to 850 DPH_{300}.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 3% max. Unmelted Particles - 10% max.

Deposit Efficiency and Deposition Rate

60% min for N_2/H_2 @ 5.5 lb/hr
60% min for A/H_2 @ 5.5 lb/hr

AMSC N/A FSC 3439

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Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

Aerospace Materials Specification	AMS 7875
General Electric Company	B50TF137, Class A
Pratt & Whitney Aircraft	PWA 1305
Rolls-Royce Ltd.	MSRR 9507/17

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/39)

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